# SmartOPD: Occupant-Aware Computer Vision and Fuzzy Wait-Time Estimation for Outpatient Queue Management

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***Abstract*—Outpatient departments (OPDs) in small hospitals typically issue paper tokens, announce them by voice, and offer no reliable wait estimate, which drives crowding around the reception desk. This paper presents SmartOPD, an end-to-end, privacy-preserving system that couples a commodity camera to a live digital queue. A YOLO11n detector and a lightweight centroid tracker run server-side on CPU and convert people into *occupants* of a doctor-room zone: an entry is only confirmed after a dwell period, people already inside when tracking starts become token-less *ghosts*, and exits are matched per occupant so that a patient leaving never completes somebody else's consultation. Measured consultation lengths feed an adaptive prior-to-median estimator whose output is refined by a Mamdani fuzzy inference layer that reasons over consultation length, measurement confidence, spread, queue position and the progress of the live consultation, returning a wait range with a confidence tag and a linguistic label. Because demonstration clips compress time, durations are stored in raw video seconds and converted to clinic minutes by an operator-tunable factor. Three browser interfaces (monitoring console, nurse desk, patient display) share one FastAPI state store. The implementation ships with 58 unit tests and a headless replay harness that reproduces six complete consultations in order over a frame-continuous ping-pong clip, with no token ever moving backwards and every consultation training the estimator.**

***Index Terms*—computer vision, person detection, multi-object tracking, outpatient department, queue management, fuzzy logic, wait-time prediction, edge inference**

## I. Introduction

Queueing is the dominant experience of an outpatient department (OPD). A patient receives a paper token, sits in a hall, and depends on a nurse to call numbers by voice. Two failures follow: patients cannot plan their time, so they cluster around the desk to ask for updates, and the clinic has no measurement of how long consultations actually take, so any estimate it offers is a guess. Operations-research work on outpatient scheduling has long shown that appointment rules and queue discipline materially affect waiting and idleness [1], [2], yet these models assume that service times are known or estimated from historical records.

Cameras already exist in most clinics, and modern single-stage detectors make person detection inexpensive on commodity hardware [3]–[5]. The gap addressed here is not detection accuracy but *semantics*: converting anonymous detections into queue events that a hospital can trust, and converting noisy measurements into a wait estimate a patient can act on.

This paper makes four contributions.

1) An *occupant-based* room model that links a queue token to a tracked person only after a dwell-confirmed entry, distinguishes ghost occupants from admitted patients, and completes consultations per occupant so that simultaneous entry and exit (a swap at the door) never corrupts the queue.
2) Seam handling for looped video sources: a frame-continuous ping-pong construction allows tracker identity and consultations to survive the wrap, while jump-cut sources freeze events and re-bind occupants after the cut.
3) A fuzzy wait-time estimator that replaces the brittle "remainder plus people-ahead times average" formula with a Mamdani inference system over five linguistic inputs, producing a range, a confidence tag and a patient-facing phrase.
4) A reproducible implementation: three shared-state web interfaces, 58 unit tests, and a headless replay harness that asserts ordering and no-backwards invariants on a real clip.

## II. Related Work

### A. Person detection and tracking

Single-stage detectors such as YOLO regress boxes and class scores in one pass and are the de-facto choice for real-time person detection [3], [4]; the YOLO11 family provides small weights that run on CPU [5]. Tracking is dominated by detection-based association: SORT [6] matches boxes with a Kalman filter and Hungarian assignment, DeepSORT [7] adds an appearance embedding, and ByteTrack [8] associates low-confidence boxes as well. For a single fixed camera observing a waiting room, greedy nearest-neighbour association is sufficient and avoids an appearance model, at the cost of identity switches under long occlusions.

### B. Crowd analytics in clinical spaces

CNN-based crowd counting and density estimation are mature [9], but the literature largely reports head-count accuracy on public benchmarks. Clinical queue management

TABLE I
THE THREE INTERFACES AND THEIR USERS

| Interface | User | Function |
|---|---|---|
| Monitoring console | operator | live annotated feed, zone drawing, crowd counters, reason-coded event log |
| Nurse desk | nurse | registration, token issue, call/skip/no-show, undo, time-factor slider, CSV export |
| Patient display | waiting patients | now-serving board, next-up token, per-token wait range and confidence |

additionally needs *door events* (who entered, who left) and a mapping from events to named tokens, which is where the occupant model of Section IV contributes.

### C. Outpatient scheduling and queueing

Reviews of outpatient scheduling [1], [2] optimise appointment intervals, panel sizes and overbooking under stochastic service times; classical queueing theory [10] supplies the wait expressions that our crisp baseline approximates. These models assume service time distributions are known; SmartOPD measures them from the camera and feeds them back, which closes the loop between operations research and computer vision.

### D. Fuzzy decision support

Fuzzy sets [11] and Mamdani inference [12] remain attractive when inputs are imprecise and stakeholders need interpretable rules [13]. Waiting-time communication is exactly such a setting: "about ten minutes, medium confidence" is more honest than a false-precision single number.

## III. System Architecture

Fig. 1 shows the architecture. A single Python process serves both the vision pipeline and the web application. A background worker thread reads frames, runs detection and tracking, and updates the queue engine; the FastAPI layer [15] exposes JSON endpoints and three pages. State is persisted atomically to a local JSON file every two seconds, so a crash cannot corrupt the queue. Frames exist only in memory; only anonymous counters, tokens and durations are stored. No cloud service is contacted after installation.

## IV. Vision Pipeline

### A. Detection

Every processed frame is resized to 640 px and passed to YOLO11n restricted to the `person` class at a confidence threshold of 0.35 [5]. On commodity CPU hardware the pipeline sustains 10–16 processed frames per second, which is ample for room-level queue semantics; frames are deliberately not processed at the source rate.

### B. Tracking

Detections are associated to existing tracks by greedy nearest-neighbour matching on the foot point $(c_x, c_y)$ of each box,

$$\hat{j} = \underset{j:\ \|c_i - d_j\|_2 < d_{\max}}{\arg\min} \|c_i - d_j\|_2, \tag{1}$$

with $d_{\max} = 0.18$ in normalized coordinates. A track survives 15 consecutive misses ($\approx$ 1.5 s) before it expires, which absorbs short occlusions; expiry is reported to the analyzer as a potential door event.

### C. Occupant model

A token is a human artifact; a track is a machine artifact. The analyzer therefore maintains *occupants*, $\mathcal{O} = \{(k, \tau, t_{\text{in}})\}$, where $k$ is a track key, $\tau$ the linked token (or $\varnothing$) and $t_{\text{in}}$ the time the person was first seen inside the doctor zone. Three rules make the mapping robust.

**Entry requires a sighting outside and a dwell.** A track seen in the waiting area that moves inside the doctor zone is confirmed only after $\Delta_{\text{entry}} = 2$ s of continuous presence, i.e.

$$\text{entry}(k) \iff t_{\text{in}} \text{ recorded } \wedge\ (t_{\text{now}} - t_{\text{in}}) \geq \Delta_{\text{entry}}. \tag{2}$$

Shorter crossings are logged as pass-bys and touch no token. A track that is inside at tracking start (or after a seam) and was never seen outside becomes a *ghost* occupant with $\tau = \varnothing$: it occupies the room but cannot consume a token.

**Exit is per occupant.** An occupant leaves when its foot point stays outside the zone for four consecutive frames (single-frame flicker cannot end a consultation) or when its track vanishes at the door. Only the leaver's token is completed. In a swap—one patient walking out while another walks in—the two events are independent and the queue advances exactly once.

**Tokens never move backwards.** The camera may transition a token waiting $\rightarrow$ called $\rightarrow$ in $\rightarrow$ done only. Earlier prototypes reverted an admission when a stay looked too short; on short demonstration clips that rule oscillated between states and is removed. Noise protection now happens strictly before admission.

### D. Seams and continuous loops

Demonstration clips are seconds long while consultations are minutes long. Two mechanisms bridge the gap. First, a *ping-pong* source is generated by concatenating the clip with its reverse; every join and the wrap are frame-continuous, so the tracker is *not* reset and a consultation may span several loops. Second, ordinary (jump-cut) sources trigger a *seam*: events are frozen for $\Delta_{\text{hold}} = 1.5$ s, the tracker is reset, and occupants are re-bound to the people the camera then sees (Algorithm 1).

### E. Crowd analytics

Two counters are derived per frame: *in frame* (all live tracks) and *in waiting area* (stable tracks, i.e. at least two hits, whose foot point lies in the waiting zone *and not* in the doctor zone). Excluding the doctor zone is what makes the public number

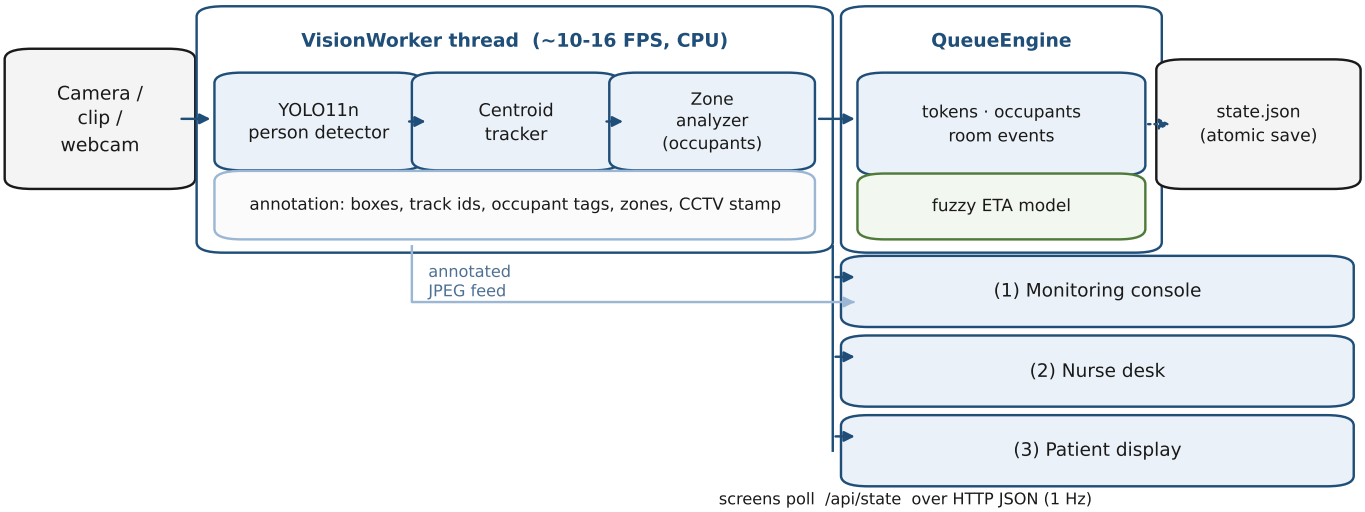



Fig. 1. System architecture. One server process hosts the vision worker (YOLO11n → centroid tracker → zone analyzer), the queue engine with the fuzzy estimator, and the HTTP layer that feeds three browser interfaces. The annotated JPEG stream drives the monitoring console; all screens poll `/api/state`.

**Algorithm 1** Seam re-binding of occupants

freeze entry/exit events for $\Delta_{\text{hold}}$
reset tracker (all track keys die; occupants survive)
**after** the hold, take the set $I$ of track keys inside the zone
order occupants $\mathcal{O}$: token-bearing first, then by $t_{\text{in}}$
**for** $i \leftarrow 1$ **to** $\min(|I|, |\mathcal{O}|)$ **do**
    re-key $\mathcal{O}_i$ to $I_i$; reset its miss counter
**end for**
every remaining key in $I$ becomes a new ghost occupant
occupants without a visible track are *kept* (never guessed away)
log REBOUND with counts (re-bound, new ghosts, kept)

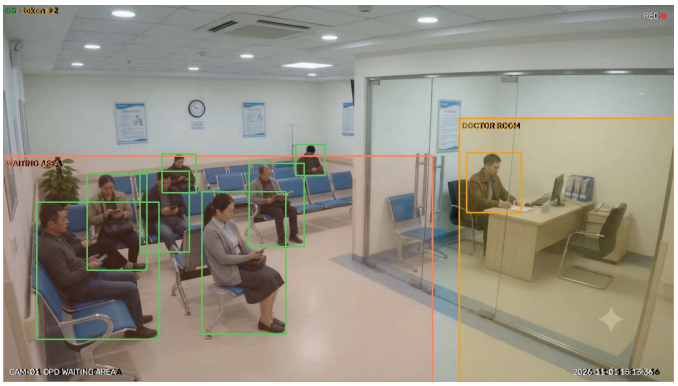



Fig. 2. Annotated frame produced by the server. Waiting and doctor zones are tinted; each person carries a track id and, inside the doctor zone, its occupant tag (a token number or `ghost`). A CCTV-style label, timestamp and recording dot make the stream readable on a wall display.

fall when a patient is admitted instead of counting the same person twice.

## V. Queue Semantics

A token moves through $\{\text{waiting}, \text{called}, \text{in}, \text{done}\}$ with $\{\text{skipped}, \text{noshow}\}$ reachable from waiting/called and re-queueable. When an entry is confirmed, the engine links a token in a fixed priority order: (i) a token the nurse already *called*; (ii) otherwise, if the camera-advance mode is enabled (default), the next waiting token by priority flag and registration order; (iii) otherwise the occupant is recorded as unlinked. The nurse can always override with manual enter/free, skip, no-show, re-queue, and an *undo* that returns the most recent completed token to the queue and removes the corresponding measurement.

Every transition is written to an event log with a reason code (ENTRY_CONFIRMED, PASS_BY, EXIT_LINKED, EXIT_GHOST, SEAM_HOLD, REBOUND) and a source (`camera`, `nurse`, `manual`, `simulation`). The code makes the system auditable: an operator can distinguish "the camera admitted a patient" from "the nurse corrected a mistake".

## VI. Fuzzy Wait-Time Estimation

### A. Learned consultation length

Let $t^{(1)}, \ldots, t^{(n)}$ be the raw durations of the last $W = 12$ consultations, measured in video seconds. They are converted to clinic minutes by the factor $\kappa$ (default $\kappa = 25$; $\kappa = 1$ for a live webcam) and blended with an 8-minute prior $P$,

$$\hat{m} = (1 - w)\, P + w\, \text{median}_{i \le W} \left( \frac{\kappa\, t^{(i)}}{60} \right), \qquad (3)$$

with blending weight $w = \min(1, n/4)$. Storing raw seconds (rather than pre-scaled minutes) means the operator can slide $\kappa \in [10, 120]$ and every historical measurement is re-scaled consistently, including ETA bounds already on screen.

TABLE II
FUZZY RULE BASE (ANTECEDENT → CONSEQUENT)

| Antecedent | Consequent |
|---|---|
| length short ∧ ahead none | wait tiny |
| length short ∧ ahead few | wait small |
| length short ∧ ahead many | wait medium |
| length medium ∧ ahead none | wait small |
| length medium ∧ ahead few | wait medium |
| length medium ∧ ahead many | wait large |
| length long ∧ ahead none | wait medium |
| length long ∧ ahead few | wait large |
| length long ∧ ahead many | wait huge |
| progress ending ∧ ahead none | wait tiny |
| confidence low | wait medium (weight 0.6) |

### B. Membership functions

Five inputs are fuzzified with triangular and trapezoidal sets (Fig. 3): consultation length (short/medium/long), measurement confidence from $n$ (low/medium/high), spread via the coefficient of variation (tight/moderate/wide), queue position (none/few/many) and progress $p = (t_{\text{now}} - t_{\text{in}})/\bar{t}$ of the live consultation (just started / mid / likely ending).

### C. Rule base and defuzzification

Table II lists the rule base. Rule strength uses the minimum of antecedents, rules are aggregated by maximum, and the output is defuzzified by the centroid method over a discretized universe $X = \{0, 0.5, \ldots, 120\}$ clinic minutes,

$$y^* = \frac{\sum_{x \in X} x\,\mu(x)}{\sum_{x \in X} \mu(x)}, \qquad \mu(x) = \max_r \min\big(\alpha_r, \mu_{C_r}(x)\big). \tag{4}$$

The bounds are derived from the same fuzzy state rather than a fixed percentage,

$$s = 0.18 + 0.32\,\mu_{\text{wide}} + 0.22\,(1 - \mu_{\text{high}}), \tag{5}$$

$$[y^-, y^+] = \big[y^*(1 - s),\ y^*(1 + 1.15\,s)\big], \tag{6}$$

so sparse or erratic measurements widen the interval instead of pretending precision. A consultation that has already overrun its average activates "ending", which pulls the first waiting patient towards *almost your turn*. Fig. 4 shows the resulting surface: the wait grows with queue position and with the learned length, with the highest sensitivity for long consultations.

### D. Presentation

Each waiting token is exposed as a range, a confidence tag and a phrase. The nurse sees e.g. "5–19 min, low confidence"; the patient display shows "about 5–18 min" together with the expected clock time and the number of patients ahead. The phrase is deliberately qualitative where the model is uncertain, which we consider a feature rather than a limitation.

TABLE III
TEST SUITES AND WHAT THEY PIN DOWN

| File | # | Coverage |
|---|---|---|
| `test_queue.py` | 31 | token lifecycle, priority order, occupant admission, swap safety, ghost exits, never-backwards invariant, undo, serialisation and migration, CSV |
| `test_vision.py` | 17 | tracker identity and expiry, entry dwell, pass-by, flicker, vanish exits, seam hold and re-binding, continuous-loop identity, crowd counting |
| `test_fuzzy.py` | 10 | monotonicity in queue position and length, progress sensitivity, confidence widening, prior fallback, label buckets |

## VII. INTERFACES

All three pages poll `/api/state` once per second, so any screen can run on any device on the local network.

**Monitoring console** (Fig. 5) shows the annotated feed, the two crowd counters, the occupant list of the doctor room and the reason-coded event log, and offers zone drawing (persisted per clip), a conservative auto-detect that only proposes a door near a frame edge, manual fallback buttons and a simulation for camera-less dry runs.

**Nurse desk** (Fig. 6) is the operational surface: registration with priority flag, the live queue with a NEXT marker, call / skip / no-show / re-queue, undo of the last consultation, CSV export, the camera-advance toggle and the demo time-factor slider with live re-scaling.

**Patient display** (Fig. 7) is a kiosk board: now serving, next up (flashing "please proceed" when called), waiting count, and a per-token lookup giving patients ahead, a range, a phrase, the expected clock time and the number of consultations the estimate is based on.

## VIII. EVALUATION

### A. Unit tests

The suite contains 58 tests (Table III) and runs in under a second on CPU. Vision tests drive an injected clock, so time-based rules are tested deterministically rather than by sleeping. Property-style tests assert that more people ahead never shorten a wait and that sparse data never narrows the bounds, which protects the estimator against future refactoring.

### B. Headless replay

A replay harness (`scripts/replay_check.py`) runs the real detector, tracker, analyzer and engine over the ping-pong clip for three loops with six tokens waiting, using video time as the engine clock. It asserts that all six tokens complete, that completions occur in registration order, that no token ever moves from `in` back to `called`/`waiting`, and that every completed consultation trains the estimator.

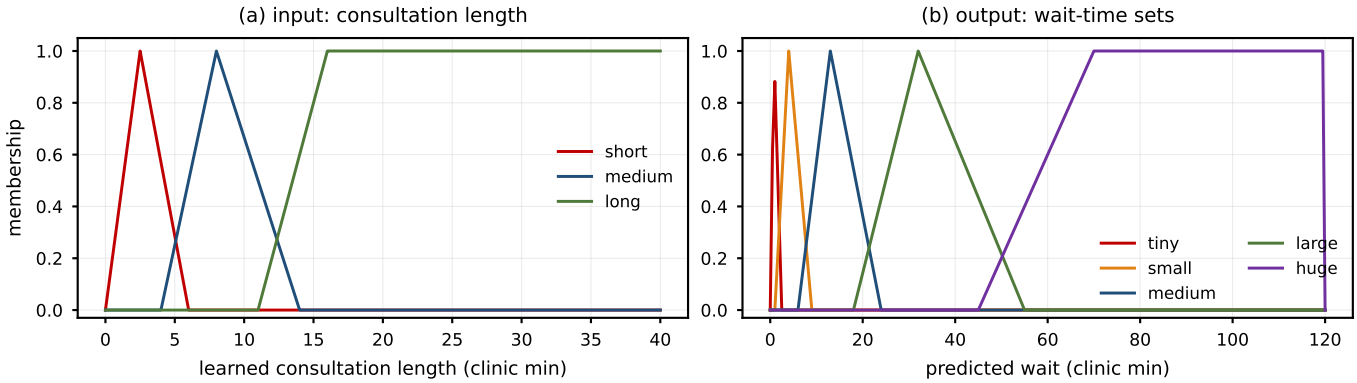



Fig. 3. Membership functions of the fuzzy estimator. (a) The learned consultation length is partitioned into short, medium and long sets. (b) The defuzzified wait is expressed over tiny, small, medium, large and huge output sets.

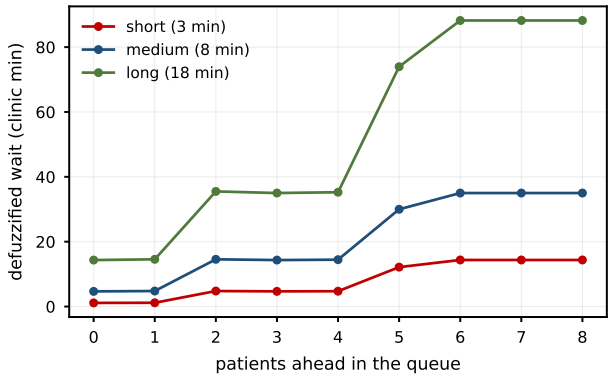



Fig. 4. Defuzzified wait versus queue position for three learned consultation lengths (8 samples, moderate spread, consultation at 30 % of its expected length).

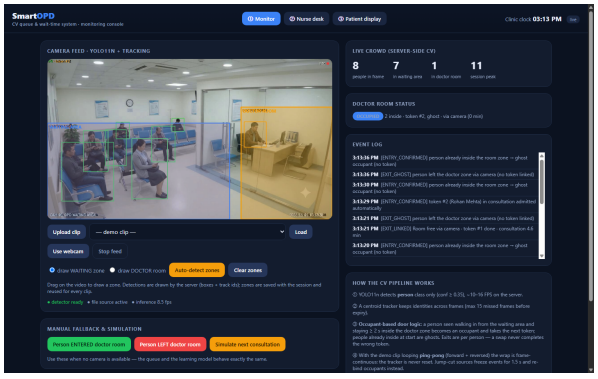


Fig. 5. Monitoring console: annotated CCTV feed with zones and occupant tags, live crowd counters, room occupants, and the reason-coded event log.

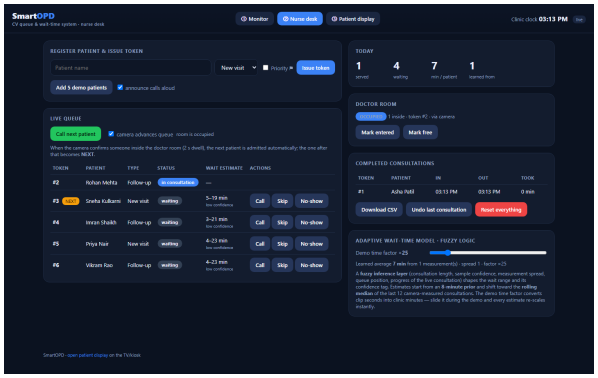


Fig. 6. Nurse desk: queue with NEXT marker, fuzzy wait ranges with confidence tags, doctor-room occupants, completed consultations with undo, and the demo time-factor slider.

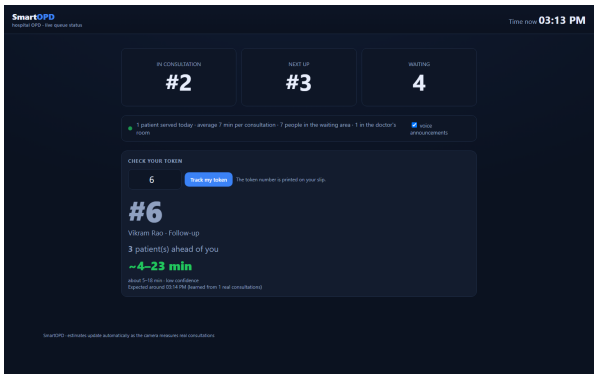



Fig. 7. Patient display: now-serving / next-up / waiting board and a per-token estimate with a linguistic label and confidence.

The alternating durations are a property of the clip geometry: the room is entered from one side and left from the other, so the forward and reverse halves alternate long and short stays. This is exactly the case that broke the earlier "short stay implies revert" rule; under the current semantics both durations are legitimate consultations and the queue advances on each exit (Table IV). The replay also demonstrates the intended invariances: the woman who is already inside at $t = 0$ is admitted as a ghost and her exit is logged EXIT_GHOST without touching a token, while the man who walks in consumes token #1.

### C. Live operation

On the demo clip the server reports 15–19 persons in frame at 10–16 processed FPS on CPU, with two occupants

TABLE IV
REPLAY RESULTS (SIX TOKENS, PING-PONG CLIP, $\kappa = 25$)

| Token | Status | Raw $t$ (s) | Clinic (min) | Learned |
|---|---|---|---|---|
| #1 | done | 11.0 | 4.6 | yes |
| #2 | done | 4.6 | 1.9 | yes |
| #3 | done | 11.0 | 4.6 | yes |
| #4 | done | 4.6 | 1.9 | yes |
| #5 | done | 11.0 | 4.6 | yes |
| #6 | done | 4.6 | 1.9 | yes |
| completed / learned | | 6/6 | 6/6 | |

inside the doctor zone during a swap. After a single measured consultation the estimator shows the expected behaviour of a low-confidence state: wide ranges (e.g. 5–19, 3–21 and 4–23 minutes for successive queue positions) tagged *low confidence*. Moving the factor from $\times 25$ to $\times 120$ re-scales the same measurements immediately, which operators use to match the pace of a staged demonstration.

### D. *What is* not *measured*

We deliberately do not report precision/recall for door events or mean absolute error for ETAs: the project has no frame-level ground truth annotation, and inventing one from the same detector would be circular. A proper field study needs a hand-labelled subset of entry/exit events ($F_1$, latency) and a log of realised consultation times (MAE, coverage of the predicted interval). Section IX lists this as the primary next step.

## IX. LIMITATIONS AND ETHICAL CONSIDERATIONS

**Identity.** The camera does not know who a person is. The system assumes queue order: the person who walks in is the next token. This is acceptable for a single-server OPD, but a patient who jumps the queue will consume somebody else's token, and the nurse must correct it with undo. Person re-identification [14] would be required to verify identity and is left as future work.

**Single room.** One doctor zone, one active queue. Multi-room operation requires per-zone occupants and per-room queues.

**Visibility.** The patient must remain visible inside the zone while consulting; if the room is fully out of view, the nurse falls back to manual enter/free. Occlusion at the door can delay an exit by up to the miss budget.

**Short clips and time scaling.** The $\kappa$ factor is a calibration, not a measurement. It is exposed, logged and changeable at runtime precisely because no single value is "correct" for a staged video.

**Privacy.** Frames are processed in memory and never written to disk or transmitted; only anonymous counters, token numbers, durations and zone rectangles persist locally. A deployed system would still require signage and institutional approval, and should blur faces at the capture stage.

## X. CONCLUSION AND FUTURE WORK

SmartOPD shows that an inexpensive camera plus a disciplined event model can turn an OPD waiting hall into a measurable queue: entries and exits are attributed per person, tokens advance without nurse input, and the wait estimate becomes a range with an honest confidence tag instead of a false single number. The implementation is small, offline, auditable and covered by 58 tests plus a replay harness.

Future work follows directly from the limitations: (i) a field study with annotated door events and realised consultation times to quantify $F_1$ and ETA error; (ii) person re-identification at the door to verify the admitted patient; (iii) multi-room and multi-camera operation; (iv) patient-side QR check-in so registration and queue state share one identifier; and (v) integration with hospital information systems, for which the CSV export is the current hand-off point.

## REPRODUCIBILITY

The implementation, tests, replay harness and demo assets are released under the MIT licence.[1] The pipeline runs offline after a one-command setup that installs a local Python environment and a 6 MB detector weight file.

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